AI BASED ADAPTIVE VR MEDITATION SYSTEM

COMMON INTEGRATED SOLUTION REPORT

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Declaration and Supervisor Certification

We declare that this common report does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made. The individual and shared contributions are identified in this report and Appendix B.

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Supervisor certification. The above candidates carried out the research described in this common report under my supervision.

Supervisor name: _____

Signature: _____ Date: _____

Abstract

Personalized virtual reality meditation requires more than an immersive scene or a confident prediction. It requires reliable acquisition, bounded adaptation and evidence that links system decisions to the participant's experience. This project develops Laminar VR, an integrated prototype combining a jointly developed wearable with four individual research components: physiological stress inference, adaptive visual control, voice-based assessment and adaptive audio control.

The architecture connects BLE acquisition and mobile session management to local inference services and a Unity meditation environment. Two participant-specific contextual-bandit agents select bounded visual and audio changes using eligibility checks and delayed physiological outcomes. Independent safety limits preserve participant control. Scene- and preference-conditioned generated meditation music is separated from fixed environmental ambience and real-time playback adaptation. Voice assessment provides a complementary pre/post and cross-modal evidence pathway.

Fresh automated verification passed six relay tests and five audio-analysis tests. Component-reported physiological and voice experiments are presented with their dataset, labelling and generalization limits. These results are not combined into an unsupported overall accuracy or relaxation claim. The reviewed implementation also identifies unresolved live-input provenance, session-isolation and cross-agent attribution requirements. An integrated participant study is specified but has not yet produced results.

The common contribution is an auditable workflow that preserves uncertainty, distinguishes proposed from applied actions and excludes invalid learning evidence. The prototype establishes a substantial technical basis for controlled research, while participant benefit and production readiness remain to be validated.

Keywords: virtual reality meditation, wearable sensing, contextual bandit, multimodal assessment, adaptive audio

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We acknowledge the collaboration of group **R26-DS-006** in developing the shared wearable device and integrated workflow. Each member's individual research component contributes to a common system supported by agreed interfaces, teamwork and collective verification.

We thank the authors and maintainers of the research datasets, models, software libraries and technical documentation cited in this report. Their contributions enabled prototype development and informed our commitment to appropriate attribution, licence compliance and responsible interpretation of research findings.

Table of Contents

1 INTRODUCTION AND PROJECT CONTEXT 1

1.1 Background and Context 1

1.2 Stakeholders and User Needs 2

1.3 Alignment With the Sustainable Development Goals 4

1.4 Purpose and Scope of the Common Report 4

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 6

2.1 Overall Problem Definition 6

2.2 Root Causes and Overall Research Gap 6

2.3 Subproblem and Component Mapping 7

2.4 Overall Aim and Group Objectives 9

2.5 Scope Assumptions and Constraints 10

3 OVERALL SOLUTION STRUCTURE 11

3.1 Integrated Solution Overview 11

3.2 High Level Architecture 12

3.3 Component 1 Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals under Real-Time Deployment Constraints 14

3.4 Component 2 Safety Constrained Adaptive VR Enviornment 15

3.5 Multilingual Voice-Based Stress Assessment with Cross-Modal Session Validation 16

3.6 Component 4 Confidence Aware Personalized Adaptive Audio Control 17

3.7 End to End User and Data Flow 17

4 COMPONENT INTEGRATION AND COHESIVE OPERATION 20

4.1 Integration Strategy 20

4.2 Interfaces Data Contracts and Shared Models 22

4.3 Shared Nonfunctional Requirements 24

4.4 Integrated Workflow and Exception Handling 26

4.5 Team Contribution and Dependency Matrix 28

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 29

5.1 Common Technology Stack and Infrastructure 29

5.2 Development and Integration Process 30

5.3 Deployment and Operational Configuration 31

5.4 Evidence of Team Execution 34

6 INTEGRATED TESTING EVALUATION AND VALIDATION 35
6.1 Evaluation Framework and Success Criteria 35
6.2 Integration and End to End Testing 36
6.3 Overall Results 38
6.4 Stakeholder and User Validation 41
6.5 Achievement of Common Objectives 43
6.6 Integrated Limitations and Threats 44
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION . 46
7.1 Value Proposition and Beneficiaries 46
7.2 Market Alternatives and Differentiation 46
7.3 Business Model and Feasibility 47
7.4 Adoption Intellectual Property and Compliance 48
7.5 Sustainable Development Goal Contribution 49
8 CONCLUSION AND FUTURE DIRECTIONS 50
8.1 Overall Conclusion 50
8.2 Principal Integrated Contribution 50
8.3 Future Directions 51
REFERENCES 52
APPENDIX A INTEGRATED INTERFACE SPECIFICATION 56
A.1 Acquisition and Physiological Envelope 56
A.2 Session and Evidence Contracts 57
APPENDIX B TEAM CONTRIBUTION DECLARATION 59
APPENDIX C PROPOSED PARTICIPANT STUDY RECORD 60
C.1 Session Preparation and Stopping 60
C.2 Suggested Feedback and Analysis Record 60
APPENDIX D REPRODUCIBILITY AND SOURCE REGISTER 61
D.1 Evidence Provenance 61
D.2 Experiment Manifest Requirements 62
APPENDIX E RESEARCH AND PRODUCTION READINESS GATES 63
APPENDIX F ASSISTANCE AND AUTHOR VERIFICATION 64

List of Figures

Figure 3.1 Integrated system architecture and evidence boundaries 12

Figure 3.2 Shared wearable and mobile acquisition pathway..... 13

Figure 3.3 Session preparation and research phase sequence..... 18

Figure 4.1 Eligibility controlled online learning and independent safety filtering..... 21

Figure 4.2 Session identity and evidence linkage across components..... 24

Figure 5.1 Development deployment and external service boundaries 33

Figure 6.1 Evaluation levels and the claims each level can support 35

Figure 6.2 Proposed controlled participant evaluation workflow 42

List of Tables

Table 1.1 Stakeholder needs and integrated requirements 3

Table 2.1 Four individual research subproblems and the shared foundation 8

Table 2.2 Group objectives and required evidence 9

Table 4.1 Principal integrated interfaces 22

Table 4.2 Reviewed adaptive control safeguards 25

Table 4.3 Integrated failure and recovery behaviour 27

Table 4.4 Contribution ownership and dependencies 28

Table 5.1 Integrated technology stack and rationale 29

Table 5.2 Reviewed local service allocation 32

Table 6.1 Integrated evaluation criteria and evidence 36

Table 6.2 Technical verification evidence and scope 37

Table 6.3 Component B reported experimental results by protocol 38

Table 6.4 Component D reported voice evidence 40

Table 6.5 Objective achievement at the reviewed snapshot 44

Table 7.1 Alternative categories and intended differentiation 47

Table A.1 Minimum integrated research export fields 58

Table B.1 Member identification and individual responsibility 59

Table B.2 Contribution confirmation 59

Table D.1 Principal evidence sources 61

Table E.1 Prioritized readiness gates 63

List of Abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
BLE	Bluetooth Low Energy
CCC	Concordance correlation coefficient
DSP	Digital signal processing
GATT	Generic Attribute Profile
HR	Heart rate
HRV	Heart rate variability
I2C	Inter-integrated circuit
I2S	Inter-IC sound
JSONL	JSON Lines
LinUCB	Linear upper confidence bound
LOSO	Leave-one-subject-out
ML	Machine learning
PPG	Photoplethysmography
PPO	Proximal policy optimization
RL	Reinforcement learning
RMSSD	Root mean square of successive differences
SAC	Soft actor-critic
SDG	Sustainable Development Goal
SDNN	Standard deviation of normal-to-normal intervals

Abbreviation	Meaning
TD3	Twin delayed deep deterministic policy gradient
TLS	Transport Layer Security
UTC	Coordinated Universal Time
VR	Virtual reality
WESAD	Wearable Stress and Affect Detection dataset

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

Laminar VR is an integrated research prototype that combines wearable sensing, physiological stress inference, an adaptive virtual environment, voice-based assessment and personalized audio control. The project investigates how a meditation session can respond to an individual's preferences and changing state without treating uncertain measurements as unquestionable facts. The common contribution is the coordinated workflow across these functions, rather than any single prediction model.

Virtual reality can provide an immersive setting in which distracting visual and auditory stimuli are controlled. A randomized university-student study of mindfulness programmes with and without VR support provides a relevant motivation, including evidence concerning adherence. It does not establish that every adaptive VR system improves health or that additional adaptation is necessarily beneficial [1]. Accordingly, this project treats immersion as an enabling medium and evaluates adaptation as a separate research question.

Audio is also more than a background asset. Music and environmental sound can influence the experience of a relaxation task, but effects depend on context and the outcome being measured. Experimental work on music and stress response motivates investigating sound carefully, not assuming a universal calming soundtrack [2]. Laminar VR separates environmental ambience from personalized meditation music so that scene identity and individual listening preferences can be managed independently.

The physiological pathway uses photoplethysmography, abbreviated PPG, to derive beat timing and heart-rate-variability features. HRV is affected by respiration, posture, activity, recording quality and individual differences. Established methodological guidance therefore supports careful acquisition and interpretation rather than equating one HRV value with a diagnosis of stress [3]. The prototype uses stress-related estimates to inform bounded interaction, not to provide medical decisions.

The system also recognises that observable channels can disagree. A participant may sound tense while their cardiovascular features appear relatively settled, or an unreliable sensor may produce an apparently strong prediction. The architecture retains voice, physiology, model confidence and signal quality as separate evidence. Disagreement is recorded for interpretation rather than concealed by averaging everything into one score.

Two online controllers personalize different parts of the experience. Component C adjusts visual scene parameters. Component E adjusts audio playback parameters. Both use safety-constrained contextual bandits with delayed outcome qualification. These are reinforcement-learning-related decision systems, but they do not constitute unrestricted online deep reinforcement learning or learn from every arriving sample [7], [16], [17].

This thesis presents the integrated design and implementation as a common project report. It consolidates component-reported experiments and fresh integration tests while distinguishing them from uncompleted participant evaluation. The main conclusion is that the project has a substantial, auditable prototype architecture, but clinical effectiveness, production readiness and complete physical-device interoperability remain separate validation obligations.

1.2 Stakeholders and User Needs

The primary user is an adult participant taking part in a supervised VR meditation research session. Their immediate needs are a comfortable environment, understandable consent, appropriate audio, minimal distraction and an accessible way to pause or stop. A participant should not need to understand model probabilities or bandit mathematics to retain control of the session.

Researchers need interpretable records of what was measured, what the system decided and what actually changed. A useful log must distinguish a proposed action from an applied action, and a missing physiological outcome from an outcome of zero. Researchers also need repeatable experimental conditions so that adaptation can be compared with a strong personalized static baseline.

Supervisors and assessors require traceable individual contributions and a defensible common contribution. Component A, the wearable, is a shared engineering foundation developed by all four members. Components B, C, D and E address four individual research subproblems. This distinction prevents the hardware from being incorrectly counted as a fifth individual contribution.

Table 1.1 translates stakeholder needs into integrated requirements. Detailed preferences are intentionally separated from raw engine settings. The phone collects understandable choices, while Unity owns the mapping to supported scene and audio parameters.

Table 1.1 Stakeholder needs and integrated requirements

Stakeholder	Need	System response
Participant	Comfort and agency	Consent, pause, stop and bounded transitions
Participant	Personalization	Stable preferences plus temporary session context
Researcher	Reliable interpretation	Quality, confidence and provenance retained separately
Researcher	Reproducible comparisons	Declared conditions, frozen evaluation and versioned logs
Project team	Independent development	Versioned interfaces and component ownership
Institution	Responsible research	Approval, privacy review and evidence-bounded claims

Accessibility remains a design responsibility. Participants can differ in hearing sensitivity, motion sensitivity, language preference and familiarity with VR. The current prototype supports a constrained scene and a limited interaction flow. A broader accessible deployment would require additional interface evaluation rather than assuming that immersion is equally comfortable for everyone.

1.3 Alignment With the Sustainable Development Goals

The project is principally aligned with Sustainable Development Goal 3, particularly the intention to promote mental health and well-being under target 3.4 [23]. Its proposed contribution is a research platform for personalized, non-diagnostic relaxation support. The project has not measured population-level health improvement or progress against a United Nations indicator.

Potential educational and innovation benefits arise from developing an interdisciplinary research system that joins embedded sensing, machine learning and immersive interaction. These are secondary contributions. They do not justify claims that the prototype has already improved access to healthcare or reduced inequalities.

Responsible sustainability also requires recognising resource use. Cloud-generated audio introduces network dependence and operating cost. A wearable introduces battery and electronic-waste considerations. Reusable scene ambience, cached generated clips where permitted and carefully managed experiment schedules can reduce unnecessary resource consumption without presenting cost reduction as measured evidence.

1.4 Purpose and Scope of the Common Report

The purpose of this report is to explain how the four research components operate as one system. It documents the problem decomposition, architecture, interfaces, implementation choices, test evidence, limitations and route to responsible deployment. Individual model derivations are included only where necessary to understand integrated behaviour.

The implementation scope is an Android-led mobile workflow, a BLE wearable, local Python services and a Unity VR application centred on the Japanese Temple Pond Garden. The audio pathway generates preference- and scene-conditioned meditation clips and then adapts playback in real time. Continuous Lyria streaming access is not established by the evidence reviewed for this draft.

The source snapshot was reviewed on 11 October 2026. Current code is the authority for runtime behaviour. Component studies provide reported results that were not independently reproduced here. Appendix D records their provenance.

Medical diagnosis, treatment recommendations, unsupervised therapeutic use and validated multi-user deployment are outside scope. Participant evaluation remains prospective. This report does not claim completed recruitment, institutional ethics approval or integrated efficacy outcomes.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

The overall problem is the absence, within this project's target workflow, of an integrated and auditable way to personalize a VR meditation experience using uncertain multimodal evidence. A fixed scene and soundtrack cannot respond to individual preferences or changing state. Conversely, an adaptive system can become distracting or unsafe if it responds too aggressively to noisy inputs.

The engineering challenge is therefore not simply to maximize model accuracy. The system must acquire usable measurements, interpret them causally, select small appropriate changes, verify that those changes occurred and record a credible outcome. The research challenge is to determine whether these choices produce a more comfortable and useful experience than simpler alternatives.

The problem has several timescales. PPG samples arrive at high frequency; physiological windows summarize many beats; visual and audio actions occur conservatively; generated music takes an asynchronous request cycle; and long-term preferences persist across sessions. A single update loop cannot safely treat these processes as if they were simultaneous.

Participant control is a further constraint. Pausing, stopping, discomfort and sensor loss must take precedence over learning. A reward signal that continues to update during interruptions would incorrectly attribute unrelated changes to the controller. The integrated design therefore makes eligibility and cancellation part of the learning process, not merely interface behaviour.

2.2 Root Causes and Overall Research Gap

The first root cause is measurement uncertainty. Motion, contact quality, timing gaps and sensor limitations can distort PPG. A confident classifier does not correct a poor input signal. The architecture must preserve acquisition quality separately from prediction confidence and stop adaptation when evidence is too weak.

The second root cause is individual variability. A model trained on one population or sensor may not transfer directly to another. WESAD is an important multimodal stress research resource, but benchmark performance does not establish zero-shot performance from the project's wrist wearable [4]. Personalization must therefore be investigated under explicit transfer and cold-start assumptions.

The third root cause is a mismatch between adaptation and evaluation. Smooth-looking changes do not establish physiological benefit. Overlapping windows are statistically dependent, and two simultaneously adapting controllers can confound one another's outcomes. These problems require time-aware logging and an experimental design that separates component effects from package-level effects.

The fourth root cause is fragmented multimodal interpretation. Speech emotion, vocal stress and cardiovascular state are related but different constructs. Emotion2vec and MELD support useful representation-learning research, yet neither makes a voice model a clinical stress detector by default [5], [6]. The project must explain the limits of each output before using disagreement as meaningful evidence.

The project's research gap is consequently specific: the design and evaluation of a preference-aware, confidence- and quality-gated VR meditation workflow with two bounded online controllers and a complementary voice assessment pathway. It is not a claim that no adaptive VR, wearable stress model, speech model or generative music system already exists.

Safe reinforcement learning literature provides a wider foundation for constrained decision-making [8]. Laminar VR contributes a concrete integration of eligibility checks, delayed attribution and independent output limits within a meditation prototype. Whether this integration improves participant outcomes remains a hypothesis requiring controlled evaluation.

2.3 Subproblem and Component Mapping

Table 2.1 maps each individual component to its research question and output. The shared wearable is an enabling input layer, not a competing research controller. Its

acquisition contract is required by B, while the phone microphone supplies D's recorded voice samples.

Table 2.1 Four individual research subproblems and the shared foundation

Part	Research subproblem	Principal output
A shared	Acquire wearable signals with traceable timing	Raw PPG and environmental noise telemetry
B	Estimate stress-related state from causal physiological context	Stress distribution, band, confidence and quality
C	Personalize bounded visual adaptation under delayed evidence	Safe scene-parameter decisions and outcomes
D	Assess vocal state and cross-modal disagreement	Voice scores, change and disagreement records
E	Personalize bounded audio control around generated content	Safe playback decisions and outcomes

Component B investigates whether participant-relative physiological context improves prediction and how much performance is retained under causal deployment. Component C investigates personalized scene control through a contextual bandit. Component D investigates voice representations, pre/post change and complementary interpretation. Component E investigates personalized audio adaptation with explicit generation and playback boundaries [14]-[19].

The common research question is how these outputs can be connected without losing their uncertainty, timing, identity or meaning. A service returning a JSON object is not sufficient integration if the receiving controller misunderstands the confidence definition or learns from an unrelated participant's latest result.

2.4 Overall Aim and Group Objectives

The overall aim is to develop and technically evaluate an integrated, preference-aware VR meditation prototype that uses physiological and vocal evidence responsibly while maintaining safe, explainable adaptation.

Table 2.2 defines the group objectives and the evidence needed to assess them. The objectives separate implementation from effectiveness. A passing interface test can establish message compatibility, but it cannot establish that a participant became calmer.

Table 2.2 Group objectives and required evidence

ID	Objective	Evidence requirement
O1	Establish a traceable sensing-to-state pipeline	Timing, quality and physical-device recordings
O2	Deliver a preference-configured VR session	Session configuration and applied-output logs
O3	Constrain visual and audio adaptation	Safety, interruption and action-attribution tests
O4	Add complementary voice assessment	Recording, inference and cross-modal validation
O5	Preserve reproducible experiment records	Conditions, versions, exports and participant-level analysis
O6	Evaluate integrated usability and benefit	Approved participant study and declared outcomes

These objectives support hypotheses rather than presupposed conclusions. A personalized static condition may already perform well. An adaptive condition may improve preference fit while adding distraction. Voice and physiology may disagree without either being erroneous. The evaluation must allow these outcomes instead of defining success only as a favourable adaptive result.

2.5 Scope Assumptions and Constraints

The current development architecture assumes one active paired participant session. This is consistent with the relay and the global latest-stress interfaces, but it is not suitable evidence of safe concurrent multi-user operation. Multi-user deployment requires identity-bound physiological subscriptions, authorization and isolation tests.

The Japanese Temple Pond Garden is the only currently selectable scene. Visual parameters are normalized before mapping to scene-specific engine settings. Fixed temple ambience belongs to the environment, whereas meditation music is generated from the participant's audio preferences and the selected scene reference.

The wearable documentation includes an MLX90614 temperature sensor, but the reviewed firmware does not implement its live reading and transmits a null temperature field. Component B can use a configured fallback. Any study must identify that fallback explicitly and must not describe synthetic temperature context as a measured wearable value [13]-[15].

The mobile voice full-session call currently requests mock HRV. This is a development configuration and must be disabled and verified before claiming a real cross-modal participant workflow. The report therefore distinguishes available code paths from a fully demonstrated live chain [20].

Cloud model access, network latency and API availability constrain generated music. The implemented clip pathway cannot guarantee instantaneous composition. The session-start requirement is instead enforced by waiting for a generated personalized clip to load before the meditation session begins [17], [22].

Safety thresholds and transition limits are research configuration, not clinically validated dose limits. Ethics approval, informed consent, equipment checks and supervised stopping procedures remain prerequisites for participant research. Commercial and clinical use would require a separate assessment.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

The integrated solution begins with participant consent and stable preferences on the mobile application. A session-specific check-in captures temporary context without silently overwriting long-term preferences. The phone connects to the wearable, prepares a paired VR session and records a pre-session voice sample when that assessment is included.

The wearable supplies raw PPG to the phone. The phone reconstructs a regular frame and sends it to Component B. B produces a stress-related estimate with confidence, quality and physiological summaries. Unity receives accepted state windows and exposes them to the visual and audio controllers under their respective eligibility policies.

The selected VR scene determines the environmental context and fixed ambient sound. Component E combines this context with audio preferences to request generated instrumental meditation music. Playback begins only after the required generated content is available. Subsequent visual and audio changes remain bounded and are logged independently of content generation.

Figure 3.1 shows the system boundaries. The voice pathway is complementary to the runtime physiological pathway. Supabase stores authenticated application records, while raw PPG transport and high-frequency controller events use separate channels.

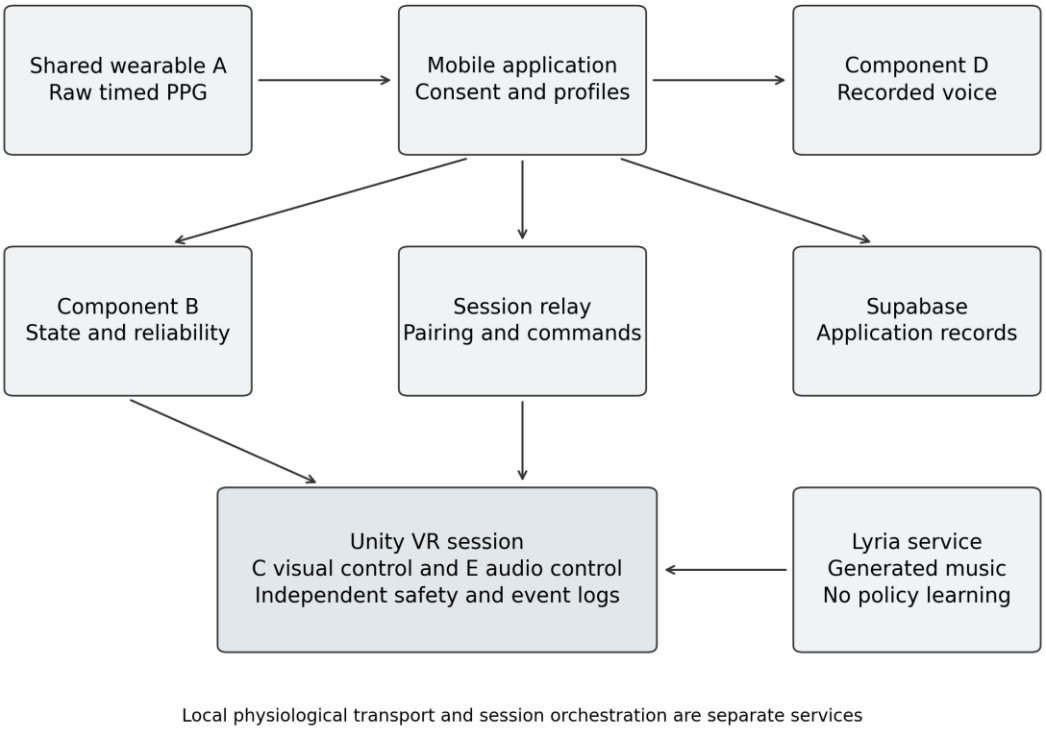


Figure 3.1 Integrated system architecture and evidence boundaries

The solution is deliberately modular. Model inference, session orchestration, audio generation and Unity playback are different responsibilities. This makes it possible to test them independently and to degrade safely when one optional service is unavailable.

3.2 High Level Architecture

The acquisition boundary consists of the jointly developed ESP32-S3 wearable and BLE communication. The wearable documentation describes a wrist-oriented prototype with a MAX30100 PPG sensor, an INMP441 microphone and an MLX90614 infrared temperature sensor. Espressif and sensor documentation support the choice of embedded interfaces, but they do not certify the assembled prototype as a medical device [10]-[13].

The firmware samples PPG at approximately 100 Hz. It sends raw packets with five timestamped infrared samples and a separate approximately 5 Hz telemetry record containing infrared, red, environmental noise and status information. The telemetry

values are not themselves a clinically validated heart-rate or oxygen-saturation measurement. Environmental noise is an uncalibrated digital-amplitude summary, not sound-pressure level in decibels.

The phone acts as the acquisition and participant-interaction bridge. Raw PPG is resampled to 64 Hz and grouped into 15-second frames containing 960 finite values. Component B receives these frames through a WebSocket ingestion endpoint. The 5 Hz dashboard telemetry is not duplicated to fabricate a high-rate waveform [15], [20].

Figure 3.2 distinguishes the documented wearable hardware from the implemented data contract. Temperature is shown as a documented hardware capability with a current null-value boundary. This distinction matters because a diagram that depicts every sensor as live would overstate the present end-to-end evidence.

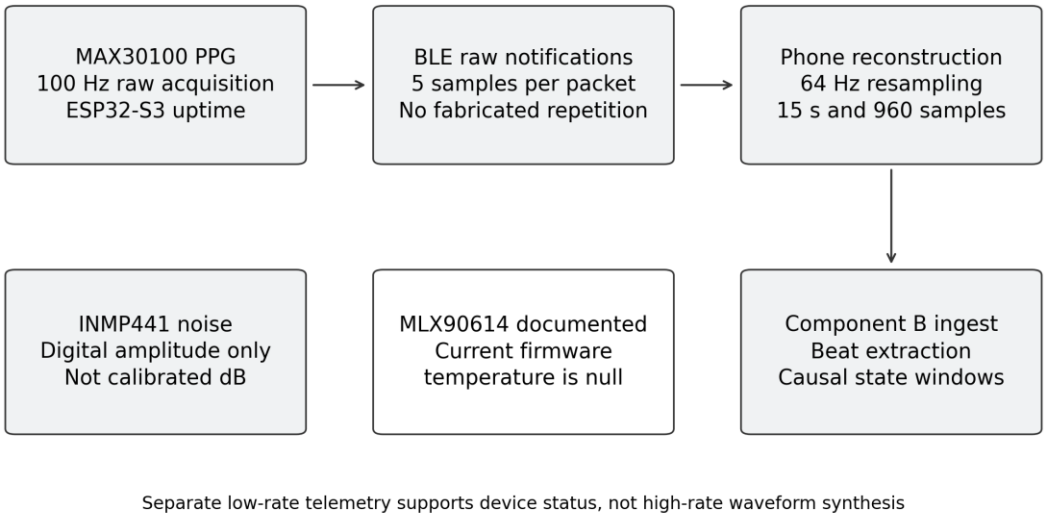


Figure 3.2 Shared wearable and mobile acquisition pathway

The orchestration boundary is the session relay. It pairs the phone and Quest, handles session commands and maintains a durable event trail. It is not the physiological model and does not replace Component B's inference service. The development relay allows one active participant session, uses a short-lived one-time pairing code and protects terminal log acknowledgement against stale or repeated requests [21].

The adaptation boundary is Unity. It owns actual scene and audio state, participant interruption and safe application of targets. The learning agents receive accepted context, propose eligible changes and wait for qualified evidence. The external Lyria backend owns music-generation requests, not the contextual bandit policy.

3.3 Component 1 Cosinor-Enhanced Multi-level Stress Inference from Wearable RR Intervals under Real-Time Deployment Constraints

Component B transforms PPG-derived beat intervals into a causal stress-related estimate. The deployed configuration combines an engineered-feature XGBoost branch and a sequence model with causal multi-scale context. The reviewed model configuration assigns weights of 0.15 and 0.85 respectively. These values describe the exported configuration, not a general recommendation for every dataset [14], [15].

Its window is 60 beats with a five-beat step. This is different from a fixed 60-second window. At 80 beats per minute, 60 beats represent roughly 45 seconds. Unity's approximately 60-second forwarding cadence is a separate delivery policy. Keeping these meanings distinct prevents incorrect reward timing and misleading descriptions of sample count.

The output distribution covers relaxed, mild, moderate and high categories. A confident estimate may be represented as a point, while an uncertain estimate can be represented as an adjacent band. Confidence is based on the separation between the two highest class probabilities. Signal quality separately represents usable physiological evidence. Neither field is the BLE signal-strength reading.

The continuous stress score is an expected ordinal value derived from the class probabilities. It is useful as a smooth control input, but it does not erase the categorical uncertainty or become a validated clinical scale. HR, RMSSD and SDNN remain accompanying measurements with their units and window boundaries.

The component's research studies investigate relative baseline and deviation information, causal endpoint labelling and deployment trade-offs. Their results are

summarized in Chapter 6. The inference service should be interpreted as providing uncertain research evidence to controllers, not issuing a medical diagnosis.

3.4 Component 2 Safety Constrained Adaptive VR Enviornment

Component C controls five normalized visual dimensions: illumination, warmth, atmospheric softness, colour richness and ambient motion. A scene adapter maps these values to lighting, fog, colour and motion settings supported by the Japanese Temple Pond Garden. Normalization prevents the mobile preference model from becoming coupled to raw Unity engine values [16].

Three whole-controller modes support comparison: Static Personalized, Rule-Based Adaptive and Contextual Bandit. Each mode shares the same scene, participant-preference initialization and safety boundary. This design allows the study to compare the adaptive decision process rather than comparing unrelated content or starting states.

The learned mode is a participant-specific disjoint LinUCB contextual bandit. It selects from 11 actions: No Change and positive or negative movement along one visual dimension. Its reviewed feature contract contains 24 elements. The action model is deliberately smaller and more interpretable than unrestricted continuous scene editing.

Learning is delayed. A selected action must be applied, transition and settle before a complete eligible physiological window can support an update. Pausing, aborting, stale data and unsafe context interrupt the process. The recorded reward includes stress-related change, optional physiological terms and action or discomfort penalties.

The design supports online personalization but does not establish that every action caused its observed outcome. Respiration, time, expectancy and concurrent audio can influence the same measurements. Controlled comparisons and explicit contamination handling are therefore required in addition to the bandit implementation.

3.5 Multilingual Voice-Based Stress Assessment with Cross-Modal Session Validation

Component D provides a recorded-voice assessment pathway before and after a session. It separates recording quality, acoustic inference, change analysis, cross-modal interpretation and anomaly analysis. This layered design makes the source of a conclusion inspectable instead of presenting one opaque score [18], [19].

The model combines a frozen emotion2vec representation with prosodic features through a trainable fusion head. The implementation uses a 1024-element speech embedding and 21 prosodic features. Freezing the large encoder is a practical choice under limited local labelled data. It concentrates learning on the smaller downstream head rather than implying that a large multilingual foundation model was trained by the project.

The current operational stress proxy is based mainly on negative valence. Its normalized form clips negative valence into a non-negative score, and its display scale multiplies that value by ten. Arousal is available descriptively but is not the principal contributor in the reviewed default configuration. Confidence based on valence magnitude is a heuristic, not a calibrated probability of correctness.

The voice pathway compares pre- and post-session recordings, and it can compare voice interpretation with B's latest physiological estimate. A disagreement label is a research interpretation category, not a diagnosis of concealed distress. Anomaly detection similarly identifies unusual session-feature patterns, not a medical abnormality.

The small available real-recording evaluation and an unsuccessful Sinhala fine-tuning attempt are reported transparently in Chapter 6. Tamil and broader speaker-independent multilingual performance remain unvalidated. Conversational speech recognition, language-model replies and text-to-speech are interaction services; they are not the acoustic stress model itself.

3.6 Component 4 Confidence Aware Personalized Adaptive Audio Control

Component E controls playback around generated meditation content. It is not a music generator. Its learning agent decides small changes to intensity, density, brightness, tempo, fade and music/ambient balance, while Lyria supplies the generated meditation recording [17], [22].

Scene identity is a controlled input. A Japanese temple session should retain temple-appropriate context in the generation prompt, rather than importing an unrelated environmental style solely because a historical preference mentions another setting. Instruments, mood, intensity and other audio preferences personalize the meditation layer within that context.

The online policy is a participant-specific disjoint LinUCB contextual bandit with 40 context features and 13 actions. Three whole-controller modes are available: Static Personalized, Rule-Based Adaptive and Personalized Bandit. The assigned condition selects the whole controller. Safety constraints remain independent of its decision process and are applied consistently across all modes.

The action set contains No Change and bounded adjustments to independent audio controls. Music and ambient mix are coupled so that the output remains valid. Candidate actions are rejected if the safety filter would alter their meaning. Only an applied, unmodified eligible action can receive a learning update.

Unity implements smooth playback mappings, not arbitrary waveform re-composition. For example, brightness maps to filtering, intensity to gain and tempo to a limited pitch-based proxy. Density is also a playback proxy rather than genuine separation and remixing of musical stems. These limitations are important when interpreting preference and relaxation claims.

3.7 End to End User and Data Flow

The participant first signs in, reviews consent and provides long-term preferences. A session check-in captures temporary subjective state and any permitted preference adjustment. Temporary context belongs to the current meditation-session record and should not silently replace the onboarding profile.

The phone then connects to the verified wearable service and pairs with the Quest through the relay. A session configuration carries the selected scene and effective preference values. Unity initializes personalized static targets and confirms readiness. The relay's preparation delay and the controller's acclimatization period serve different purposes and must not be counted as the same phase.

Generated meditation content must be ready before the meditation session starts. Fixed environmental ambience is selected from the scene configuration. A generation failure therefore blocks a strict generated-content start or presents an explicit operator recovery, rather than silently claiming raw fallback audio is personalized generated music.

Figure 3.3 summarizes the reviewed visual-session timing: approximately 120 seconds of acclimatization, 900 seconds of adaptive operation and 150 seconds of stabilization. These values define a research configuration of 19.5 minutes before external preparation delays, not a completed participant-study duration.

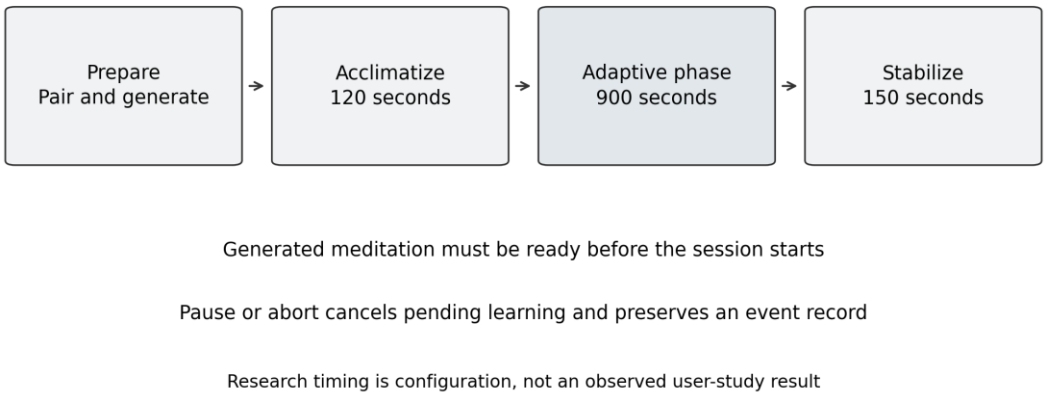


Figure 3.3 Session preparation and research phase sequence

During adaptation, accepted B windows reach the visual pathway and the audio bridge. Each controller checks its own confidence, quality, freshness, baseline and pending-outcome state. A new message does not automatically trigger a new action. Decisions and reward updates occur only when the corresponding gates permit them.

The participant may pause or stop at any time. Session completion or abortion finalizes relay records and controller logs. Post-session voice and questionnaires provide additional evidence where included. The mobile application retrieves finalized visual records and acknowledges the exact terminal export. Audio records require an explicit identity mapping or a documented timestamp-based join, rather than assuming that two local session identifiers are identical.

The resulting evidence links acquisition, physiological windows, session events, applied actions and assessments. A complete physical-device demonstration must still verify the chain and the quality of these joins.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

Integration follows an interface-first strategy. Each component retains ownership of its own model, state and output boundary. Shared contracts define what crosses a boundary: identities, timestamps, units, confidence semantics, configuration and terminal events. This is preferable to merging every service into a single process, because failures and experimental changes remain attributable.

The session relay is the orchestration plane. Component B is the physiological inference plane. Unity is the application and safety plane. Component D is an assessment plane, while the Lyria service is a content-generation plane. These planes interact, but none should be described as a universal shared backend that already performs every task.

The common learning process begins with a preference-initialized state, obtains reliable context, selects an eligible action, confirms its application and waits for delayed evidence. Both adaptive agents follow this logic. They differ in feature schemas, action sets, reward terms and output mappings, and their participant-specific model weights are not shared.

Figure 4.1 presents the shared learning pattern. The separation between proposed and applied actions is central. Without it, a controller could learn that an action was beneficial even when a safety filter prevented that action from reaching the participant.

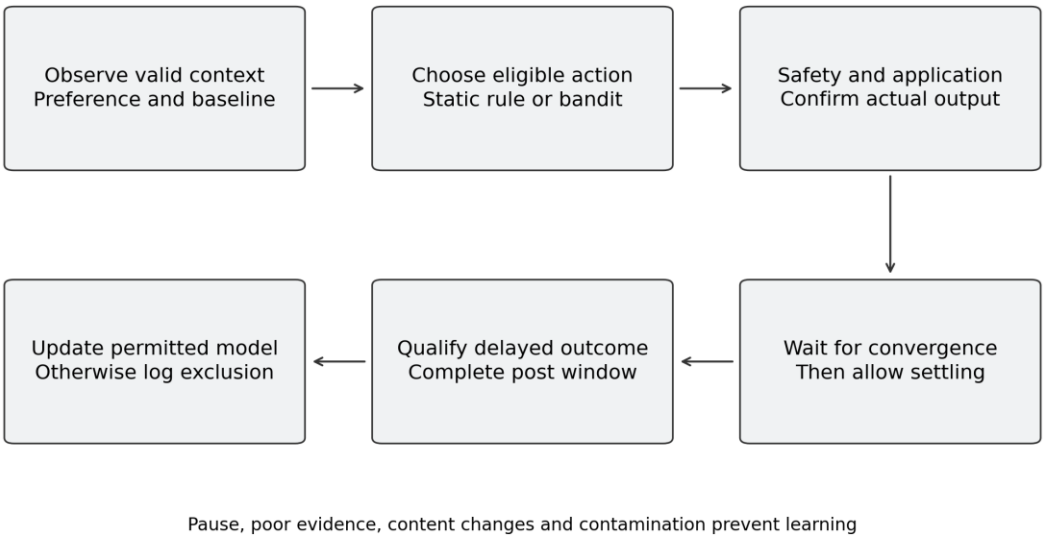


Figure 4.1 Eligibility controlled online learning and independent safety filtering

For LinUCB, each action has a regularized linear reward model. The score combines the estimated contextual reward with an uncertainty bonus. The native equations below summarize the method used as the decision-learning basis [7]. The vector x is the eligible context, A is the action's regularized design matrix, b accumulates context-weighted outcomes and alpha controls exploration.

$$\theta_a = A_a^{-1} b_a$$

$$score_a = \theta_a^T x + \alpha \sqrt{x^T A_a^{-1} x}$$

An eligible observation updates only the executed action's statistics. The initialization with a positive ridge term makes the system solvable before substantial participant history is available. Frozen evaluation disables exploratory bonuses and updates, allowing a trained policy to be assessed without silently changing during measurement.

$$A_a \leftarrow A_a + x\,x^T; \quad b_a \leftarrow b_a + r\,x$$

This choice has a practical rationale. A small discrete action library makes intervention explainable, supports masking and requires fewer observations than

unrestricted deep-policy training. However, LinUCB assumes an approximately linear contextual relationship and does not explicitly optimize long-horizon state transitions. The architecture does not claim to solve those limitations merely by calling the agent reinforcement learning.

4.2 Interfaces Data Contracts and Shared Models

Table 4.1 summarizes the principal interfaces. Detailed examples are retained in Appendix A. Endpoint availability, semantic validity and scientific validity are separate requirements. An HTTP 200 response establishes neither sensor quality nor correct participant attribution.

Table 4.1 Principal integrated interfaces

Boundary	Interface	Contract responsibility
Wearable to phone	BLE GATT raw and telemetry notifications	Device identity, count, timing and status
Phone to B	WebSocket /ingest	960 finite PPG values at 64 Hz per frame
B to consumers	WebSocket /stream	Nested stress envelope and window metadata
D to B	GET /stress/latest	Top-level latest prediction or unavailable status
Phone to Quest	Relay REST plus /realtime WebSocket	Pairing, configuration, commands and terminal logs
Unity to Lyria	POST /generate-clip	Prompt, request identity and generated audio result
Phone to D	Multipart voice endpoints	Actual recording, phase, language and session identity

The stress distribution and its derived continuous score must remain consistent. A four-category distribution is expected to contain finite, non-negative values that sum approximately to one. For category indices zero through three, the continuous score

is the sum of each index multiplied by its probability. Its range is therefore zero to three, not zero to one.

$$s = 0\,p_0 + 1\,p_1 + 2\,p_2 + 3\,p_3$$

A receiving component should not substitute the largest class probability for B's confidence margin. Nor should it substitute signal quality for confidence. The uncertainty of the model and the usability of the physiological signal answer different questions and are preserved separately in the accepted context.

Timestamp handling also requires two concepts. UTC or POSIX timestamps identify physiological window boundaries and support cross-service joins. Monotonic elapsed time controls local deadlines and transition duration. Using wall-clock time for every local timeout risks incorrect behaviour if a device clock changes during a session.

The raw wearable timestamp is an uptime value, not an absolute UTC time. The mobile acquisition layer must reconstruct and anchor that timeline, handle rollover or reset and reject unexplained gaps. A transmission containing timestamps is not automatically synchronized across devices.

The reviewed relay schema uses a session identifier, paired-device identity and typed event envelope. Its finalized log acknowledgement includes the message count and last message identifier. This permits idempotent retrieval without treating a stale acknowledgement as proof that the current terminal log has been saved [21].

Identity remains a known integration boundary. Component B creates ingestion-local processing state but exposes global stream subscribers and a global latest result. This is acceptable only under the declared one-active-participant development assumption. A production system must bind every physiological message and subscription to an authorized participant session.

Component E maintains local model and transition identities. These are not automatically the same identifiers used by the relay's finalized visual export. Figure 4.2 shows the required evidence relationship. An explicit common session key is preferable; a documented UTC join is a research workaround that requires checking for collisions and missing intervals.

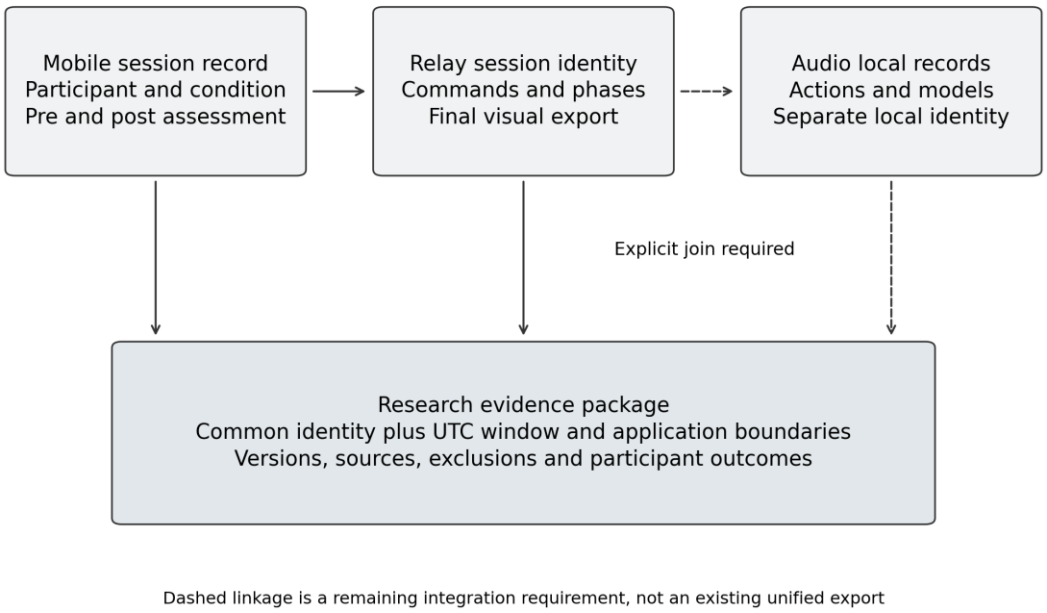


Figure 4.2 Session identity and evidence linkage across components

Versioning is part of the contract. Preference dimensions, feature ordering, reward configuration and action definitions can change. Stored online models are therefore valid only for compatible user, environment, source domain and configuration. A file loading successfully is not evidence that its feature semantics still match the current controller.

4.3 Shared Nonfunctional Requirements

Safety takes precedence over adaptation. Both controllers must retain bounded targets, smooth transitions and an explicit No Change option. Low-confidence, low-quality or stale evidence must reduce or freeze intervention. Participant pause, discomfort and emergency stopping must cancel pending learning as well as stopping visible or audible output.

Table 4.2 records the reviewed gating differences. Shared process does not mean identical thresholds. The values are provisional engineering configuration and require study-specific approval. Component E's additional confidence gate should not be incorrectly described as a feature already present in the visual context vector.

Table 4.2 Reviewed adaptive control safeguards

Safeguard	Visual component C	Audio component E
Action library	11 discrete bounded actions	13 discrete bounded actions
Context width	24 features	40 features
Decision spacing	Approximately 75 s	Approximately 75 s
Settling requirement	5 s after transition	5 s after playback convergence
Quality for normal learning	At least 0.50	At least 0.60
Confidence for normal learning	Contract-specific handling	At least 0.45
Input freshness	Approximately 90 s maximum age	Approximately 90 s maximum age
Interruptions	Cancel or withhold outcome	Cancel or withhold outcome

Reliability requires predictable degradation. Loss of an optional voice service should not corrupt an active VR controller. Loss of physiological evidence should freeze learning and retain a safe output rather than invent a low-stress reading. Loss of music generation before a strict generated-content session should prevent an invalid start.

Observability requires operational and scientific records. Operational records explain connection, timeout and load failures. Scientific records explain accepted windows, proposed actions, safety decisions, applied outputs, cancellations and rewards. A dashboard is useful for demonstration, but it cannot replace a machine-readable event trail.

Privacy requires minimization. Raw voice, derived affect, physiological traces and account identity have different sensitivity and retention needs. The phone should not upload raw PPG to general profile storage simply because that database is convenient. Authentication, row-level database policies and secure key storage are

useful controls, but do not by themselves establish complete privacy compliance [20], [25].

Performance must be assessed at each boundary. BLE packet continuity, inference delay, relay delivery, Unity frame time, audio underruns and generation latency require different measurements. One aggregate latency number would conceal which part of the participant experience is affected.

4.4 Integrated Workflow and Exception Handling

The normal workflow progresses through configuration, readiness, acclimatization, adaptation, stabilization and completion. A controller may observe several windows before it has a sufficient baseline. Baseline collection is not an excuse to fabricate optional HRV values. Missing inputs are represented explicitly and excluded from dependent reward terms.

A valid delayed outcome must begin after the intervention has actually reached the participant and after settling. Windows overlapping the action transition are not clean post-action evidence. Because B windows overlap at a five-beat step, a fresh message can still contain mostly pre-action information. Eligibility must therefore examine boundaries, not just arrival order.

Audio content changes are especially important. Loading or crossfading a new generated clip can affect the participant independently of a small playback action. Component E invalidates pending outcomes around such content changes. It also invalidates pending attribution when a visual change is observed [17].

The reviewed architecture does not demonstrate a fully symmetric shared action scheduler that isolates both agents' causal effects. Visual changes can contaminate audio outcomes, and audio changes can contaminate visual outcomes. The safe research response is to test one adaptive component while holding the other static before evaluating the complete adaptive package.

Table 4.3 defines failure responses and the evidence that should be retained. Several responses are implemented locally; complete physical-device verification remains

part of the test plan. A recovery should be visible to the operator and distinguish unavailable evidence from a valid calm state.

Table 4.3 Integrated failure and recovery behaviour

Event	Required response	Evidence to retain
Wearable unavailable	Do not claim a live physiological session	Scan, verification and connection status
PPG gap or poor quality	Reject affected evidence and freeze learning	Gap, quality and rejection reason
B unavailable or stale	Hold safe targets without fabricated values	Last valid boundary and timeout
Generation blocked or failed	Preserve explicit content state and retry deliberately	Request, model, error and outcome
Pause or discomfort	Cancel attribution and allow participant recovery	Phase, applied state and cancellation
Clip or competing-agent change	Invalidate contaminated outcome	Change boundary and affected action
Duplicate terminal export	Apply idempotent acknowledgement	Count, final ID and acknowledgement

Content moderation failures require normal error handling. Rephrasing a harmless prompt may be appropriate, but automatic repeated requests can waste resources and obscure the cause. The system must not bypass provider policies or represent a failed request as successful generation.

Online learning persistence is similarly conservative. A participant model is saved with its context and configuration identity. Simulated-source models and live-source models must not silently share state. A model update skipped because of invalid evidence is logged as skipped, not stored as a reward of zero.

4.5 Team Contribution and Dependency Matrix

The four members jointly developed the wearable foundation and the common acquisition understanding. Individual ownership then follows B, C, D and E. This report does not infer member names from unrelated files or allocate percentages without an agreed contribution declaration.

Table 4.4 records technical ownership and dependencies. Appendix B provides blank author and signature fields for the group to confirm. The matrix explains responsibility, not a verified assessment of each member's effort.

Table 4.4 Contribution ownership and dependencies

Work area	Ownership	Main dependency
Wearable design and acquisition A	All four members jointly	Verified firmware and mobile reception
Physiological inference B	Individual member B	Timed PPG and declared temperature source
Visual adaptation C	Individual member C	Preferences, accepted state and Unity scene
Voice assessment D	Individual member D	Actual phone recordings and optional B result
Audio control E	Individual member E	Preferences, scene, generation and accepted state
Common session integration	Jointly agreed integration work	Versioned contracts and shared verification

No component can claim the integrated outcome alone. B's prediction becomes useful only when the downstream controller interprets it correctly. C and E depend on shared session boundaries. D depends on recording provenance and, for cross-modal claims, a real physiological result from the same participant and phase.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

The implementation uses an embedded acquisition layer, a mobile participant layer, local inference services and a Unity presentation layer. This division supports independent testing and limits the amount of sensitive raw data that needs to cross unrelated services.

Table 5.1 records the principal stack. Versions are the reviewed project configuration rather than a recommendation to upgrade during a study. A reproducible experiment should freeze the versions actually used in the released build.

Table 5.1 Integrated technology stack and rationale

Layer	Technology	Design rationale
Wearable	ESP32-S3, Arduino, NimBLE and I2C/I2S	Accessible embedded prototyping and BLE transport
Mobile	React Native 0.87.1 and TypeScript	Participant workflow with typed application state
Mobile data	Supabase, secure key storage and local state	Authenticated records and recoverable preferences
Physiological backend	Python, FastAPI, XGBoost and neural inference	Separate causal state estimation service
Voice backend	PyTorch, emotion2vec and acoustic DSP	Frozen representation with trainable task head
Session relay	FastAPI, WebSocket and JSON Lines	Pairing, commands and durable event records
VR	Unity 6, URP, OpenXR and XR Interaction Toolkit	Scene control and headset-oriented rendering

Layer	Technology	Design rationale
Audio	Unity audio sources, filtering and Lyria bridge	Separate content generation from playback control

The reviewed Unity project uses editor 6000.0.82f1, Universal Render Pipeline 17.0.4, OpenXR 1.16.1 and XR Interaction Toolkit 3.0.11. The headset target is Quest with an ARM64 IL2CPP build configuration. An editor compilation or Play Mode test is not evidence that this configuration meets sustained headset performance requirements [16], [27].

The mobile implementation includes a native Android recording module. An older README statement that the voice port is unimplemented is inconsistent with the current Kotlin and TypeScript path. Source inspection confirms an implemented recording and request path, but does not establish a successful physical-phone recording and inference run for every device [20].

Supabase stores profile, onboarding, consent, session and questionnaire records. Raw PPG is transported to the physiological service instead of being treated as a profile field. Fine-grained controller logs remain separate and require an explicit export strategy. This reduces accidental conflation between account state and scientific evidence.

5.2 Development and Integration Process

The project follows a design-science orientation: identify the practical problem, define objectives, design and build an artefact, demonstrate it, evaluate it and communicate the findings [9]. This report uses that structure as a methodological explanation, not as evidence that all evaluation stages have already been completed.

Development decomposed the system into acquisition, inference, adaptation and assessment boundaries. Each boundary has testable contracts. For example, a valid PPG frame must contain the expected sample count and finite values; a valid reward window must satisfy time and quality constraints; and a valid terminal acknowledgement must identify the finalized log.

Integration is incremental. The wearable can be checked with a BLE inspection tool before the app is blamed for a missing device. B can be checked with a deterministic frame before Unity is involved. Unity can be exercised with explicitly simulated signals before live sensing is enabled. Each step reduces the number of simultaneous unknowns.

Simulation is retained as a separate development domain. It supports interface tests and policy experiments but cannot provide physiological evidence about participants. Component E's supplementary PPO, TD3 and SAC experiments are examples of offline simulated-user work, not the deployed online controller. PPO is a plausible deep-policy comparison method, but its existence does not make it the current runtime model [17], [26].

The online contextual-bandit choice reflects the current evidence budget. A small action library, participant-specific statistics and conservative updates are easier to inspect with sparse session outcomes. A deep policy might become appropriate with substantially more representative data, a validated simulator and controlled training. Rebuilding a neural model is not a substitute for fixing timing or attribution errors.

Generation and control are also developed independently. The Lyria backend verifies API configuration and manages asynchronous requests. Unity verifies decoding, clip assignment and crossfade behaviour. The audio agent verifies bounded playback decisions. A failure in one stage should not be explained as failure of the entire reinforcement-learning model.

5.3 Deployment and Operational Configuration

The shared repository contains a common launcher for the local demonstration services. The actual service allocation avoids using the same port for B and Lyria. Table 5.2 records the reviewed allocation. Older standalone audio instructions using port 8000 must not be copied into the integrated deployment unchanged.

Table 5.2 Reviewed local service allocation

Service	Port	Operational role
Component B	8000	Physiological ingestion, inference and latest result
Session relay	8080	Phone and Quest pairing and session exchange
Lyria backend	8002	Generated music and capability endpoints
Component D	8010	Voice and cross-modal assessment
Ollama when enabled	11434	Optional conversational language model

Figure 5.1 separates the development PC, phone, wearable, headset and external services. Loopback addresses work only within the same device. A phone or headset must use an accessible PC LAN address or an approved deployed endpoint. Firewall, network isolation and transport security must be reviewed before any sensitive session crosses a shared network.

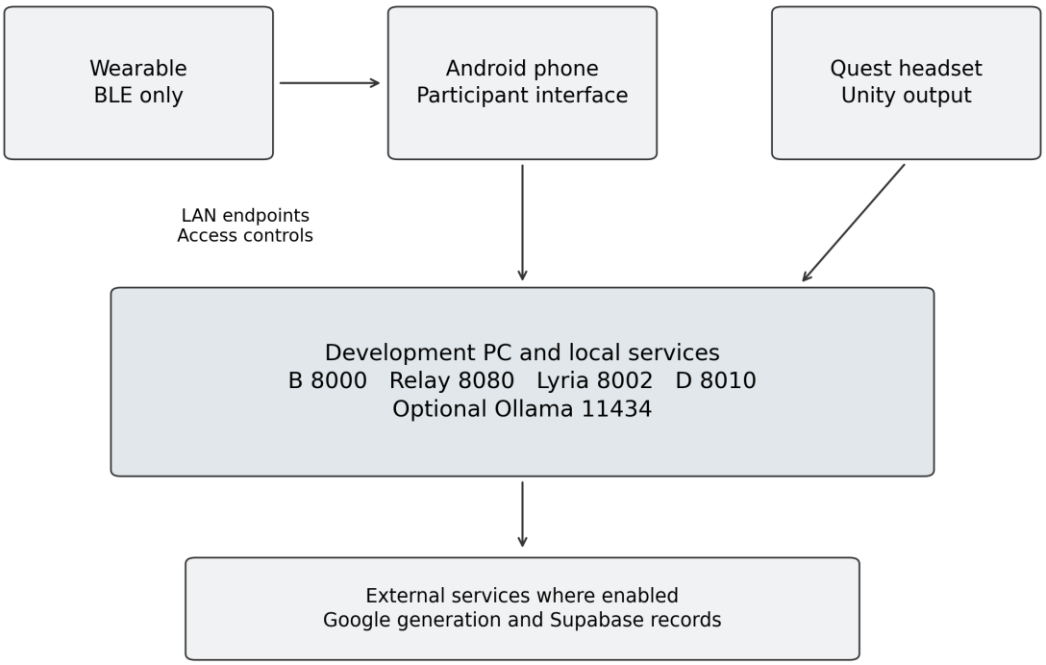


Figure 5.1 Development deployment and external service boundaries

The local launcher expects service-specific Python environments. The presence of a runtime environment under a different name does not prove the launcher will find it. A reproducibility package should record the environment path, dependency lock or requirements file, model artefact hash and command used for each service.

API keys are server-side secrets. They must not be embedded in Unity assets, mobile builds or thesis appendices. Environment examples may contain variable names, but not working keys. If a secret has entered Git history, removing it from the current file alone is insufficient; revocation and history remediation are required under the repository's security procedure.

The current Android development configuration permits cleartext local communication, and an internal release build can use a development signing setup. These settings are not production security controls. A deployable service requires TLS, managed signing, least-privilege credentials, access controls, retention rules and monitored recovery.

The relay's one-time pairing code provides a bounded device-binding workflow. It does not replace authentication for all downstream APIs. Production multi-user operation must additionally ensure that a guessed session identifier or another participant's connection cannot expose state, recordings or control commands.

5.4 Evidence of Team Execution

The repository supplies concrete implementation evidence through firmware, mobile source, model services, Unity controllers, architecture contracts and tests. The component documents explain research decisions and experiments. These artefacts support a technical execution narrative without inventing meeting attendance, supervisor approvals or participant feedback.

Fresh common-report verification executed six relay tests and five audio-evaluation tests. The relay tests cover session identity, pairing expiry, one-active-session restrictions, one-time binding, terminal acknowledgement and an in-process mobile/Quest command flow. The audio tests cover invalid outcomes, duplicate and simulated records, participant-level averaging and unavailable variability estimates.

Historical visual validation documents report 17 Play Mode passes and Edit Mode validation [27]. Earlier audio verification records describe 22 standalone C# tests and five Python tests [17]. These counts are not added together as a single integrated test score because they represent different scopes, environments and execution dates.

The reviewed scene profile still contains a false research-configuration approval flag. This is an explicit readiness item, not a cosmetic warning to ignore. Before participant use, the team must reconcile the configured scene bounds with the approved protocol and record who authorized the selected values [16].

The execution evidence therefore supports a prototype with significant implemented functionality and useful automated safeguards. It does not yet support a claim that every service is installed, every model artefact is available, the complete physical chain has passed or the system is ready for unsupervised deployment.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

Evaluation is organized into four evidence levels: implementation inspection, automated technical verification, component experiments and integrated participant evaluation. Figure 6.1 shows why these levels cannot be substituted for one another. A tested message path is necessary for a participant study, but it is not a participant outcome.

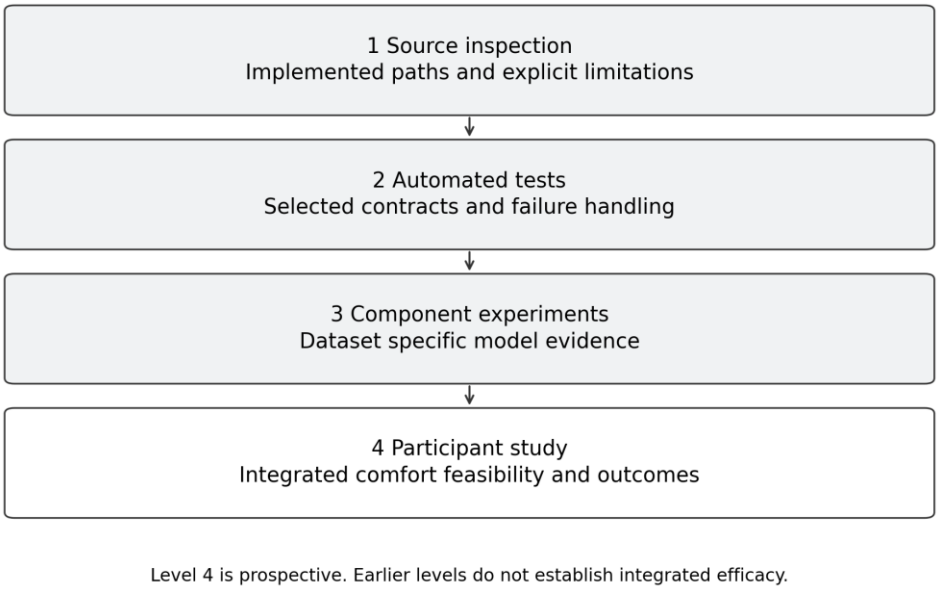


Figure 6.1 Evaluation levels and the claims each level can support

The primary integrated research objective is safe and feasible operation with valid evidence. Every intervention should be bounded, traceable and interruptible. Every learned outcome should be linked to an applied action and an eligible window. These criteria are assessable even when the study does not show a relaxation advantage.

Table 6.1 defines success criteria without fabricating thresholds after observing results. Final numerical acceptance limits for latency, headset performance and questionnaire changes must be approved before participant recruitment. A prototype requirement such as no unintended action outside bounds can be tested directly without claiming a clinical effect size.

Table 6.1 Integrated evaluation criteria and evidence

Dimension	Criterion	Measurement
Safety	No applied target outside approved bounds	Applied-state audit and fault injection
Attribution	No update from interrupted or contaminated outcome	Window and cancellation audit
Acquisition	Measured continuity with known loss and timing	Raw packet and resampled-frame trace
Usability	Participants understand control and stopping	Task observation and structured ratings
Personalization	Preferred experience without excess intervention	Preference ratings and action frequency
Relaxation support	Declared pre/post change under comparison	Self-report and qualified physiology
Reproducibility	Session can be reconstructed	Versions, conditions, timestamps and exports

A staged evaluation is recommended. First, verify the physical wearable-to-phone-to-B chain using a consenting operator under an approved engineering procedure. Second, verify one complete session with simulated inputs explicitly labelled as simulation. Third, verify a live session without adaptive learning. Only then enable online updates and compare controller conditions.

6.2 Integration and End to End Testing

The freshly executed relay suite passed six tests. These cover idempotent session creation, expired pairing-code rejection, one-active-session enforcement, one-time pairing, idempotent terminal acknowledgement and an in-process HTTP/WebSocket flow involving mobile and Quest roles. The tests ran against application test clients, not a physical phone and headset.

The freshly executed audio-evaluation suite passed five tests. It confirms that invalid outcomes are not treated as zero rewards, duplicate or legacy records are not counted twice, simulated and live domains remain separate, participant summaries receive equal weight despite unequal session counts and variability is not fabricated for one participant.

Table 6.2 records the test evidence at its actual scope. The fresh total is eleven passing automated tests. This is a verification subset, not the total number of all tests in the repository and not an eleven-session user study.

Table 6.2 Technical verification evidence and scope

Evidence	Result	Interpretation
Fresh relay suite	6 tests passed	In-process session and transport contract checks
Fresh audio analysis suite	5 tests passed	Record validity and participant-level aggregation
Historical visual validation	17 Play Mode passes reported	Component validation, not rerun for this draft
Historical audio C# validation	22 standalone tests reported	Component logic, not headset validation
Physical end-to-end session	Not established by reviewed evidence	Required before integrated live claims

Remaining end-to-end tests should include wearable restart, missing raw packets, invalid PPG values, clock jumps, network loss, B restart, stale windows, music-generation failure, decode failure, pause, discomfort, session abort and terminal-export retry. Each test should specify the expected output and whether a pending learning update must be cancelled.

The correct test result for weak physiological evidence is often no intervention. A demonstration that the system always changes something would be inconsistent with

the intended safety logic. Likewise, a test that supplies simulated temperature or mock HRV must retain that provenance in its evidence record.

An integrated smoke test should save a synchronized trace from the phone, B, relay and Unity. It should demonstrate one accepted physiological window, one safe applied action, one valid delayed outcome, one interruption and one completed export. This provides a much stronger integration artefact than screenshots of independent dashboards.

6.3 Overall Results

The available overall result is an implemented modular prototype with tested interface and analysis subsets, plus separate component research experiments. There is no defensible single overall accuracy figure. Stress classification, voice classification, action reward, packet continuity and usability measure different constructs and use different datasets.

Component B's reported mechanism study used WESAD participants under subject-held-out analysis [14]. Its HRV-only macro-F1 was 0.464. Adding time context produced 0.551, while a residual representation reached 0.646. The study interprets much of the improvement as participant-relative baseline and deviation information rather than proof of a circadian mechanism.

Table 6.3 summarizes the reported B comparisons. These values are transcribed from the supplied Component B study documentation, not newly reproduced training results. Different protocols are kept separate rather than selecting the largest number as the deployed model's performance.

Table 6.3 Component B reported experimental results by protocol

Protocol and model	Macro F1	Interpretation
Mechanism study HRV only	0.464	Baseline for the mechanism comparison
Mechanism study with time	0.551	Time-context comparison

Protocol and model	Macro F1	Interpretation
Mechanism study residual context	0.646	Relative-context result, not deployment score
Causal two-way MS-CGCA five seeds	0.5925 plus or minus 0.0129	Mean and SD reported across seeds
Bidirectional two-way five seeds	0.5993 plus or minus 0.0120	Offline comparison with future-context concern
Exported nested subject-held-out model	0.5923	Reported deployment-model estimate
Exported pooled estimate	0.6023	Less conservative than subject-held-out result

The selected causal model gives up some offline performance to avoid future information. Its reported nested subject-held-out kappa is 0.7525. This is a methodological trade-off rather than a failure to choose the numerically highest model. Endpoint labelling also matters: midpoint labels reportedly inflate F1 by roughly 0.071 to 0.084 across evaluated configurations [14].

The B labels themselves require qualification. The four ordinal categories are HRV-derived research proxies, not independent clinical diagnoses. Using related HRV information in both labels and features introduces circularity risk. A separate binary protocol-based sensitivity analysis is informative but does not remove all concerns about sensor transfer and everyday generalization.

A wrist-PPG mechanism replication is reported across 34 subjects, with residual context improving macro-F1 from approximately 0.304 to 0.355. This supports further investigation of the representation mechanism. It does not prove that the final WESAD-trained export will transfer directly to this project's MAX30100 wearable without calibration.

Component D's reported acoustic experiments show why benchmark and deployment claims must be separated [18], [19]. A frozen emotion2vec-based model outperformed the documented wav2vec2-family and hand-feature comparisons on

the acted-emotion evaluation. That result supports the representation choice, not a general claim that the system detects clinical stress accurately.

Table 6.4 preserves the voice evidence and its limitations. The real-recording set contains only 24 clips, including 11 Sinhala clips from a small speaker pool. Clip-level leave-one-out evaluation does not establish generalization to unseen speakers or to the university population.

Table 6.4 Component D reported voice evidence

Experiment	Reported result	Main limitation
wav2vec2-family comparison	60.9 percent accuracy	Acted-emotion protocol
Hand-feature comparison	70.8 percent accuracy	Different representation, same task scope
emotion2vec comparison	84.7 percent accuracy	Not a physiological stress gold standard
Real-recording clip leave-one-out	22 of 24 correct, 91.7 percent	Small sample, speaker dependence and selection bias
Sinhala head adaptation	90.9 to 81.8 percent on retained set	Negative transfer, not successful multilingual validation

The Sinhala result is an important negative finding. Retraining on newly collected Sinhala recordings did not improve the retained evaluation set. It warns against assuming that language-specific adaptation is automatically beneficial when data are sparse. Tamil performance is not established, and a confidently wrong out-of-domain example shows that the current confidence heuristic is insufficient as a correctness guarantee.

The anomaly subsystem uses a cold-start model trained with synthetic session patterns before accumulating participant history. Its unusual-pattern output must therefore be evaluated on real longitudinal sessions before being interpreted as

meaningful abnormality detection. No clinical anomaly sensitivity or specificity is claimed.

For C and E, the strongest current evidence concerns controller architecture, eligibility, safety and analysis. Neither component has completed integrated participant outcomes in the reviewed material. Their online learning behaviour is therefore described as implemented and technically testable, not empirically superior to the rule or static conditions.

Component E also retains supplementary preference-only simulations comparing rule-based control, a linear bandit, PPO, TD3 and SAC. The rule baseline had the highest synthetic composite reward, while SAC had the highest preference-match score in that exercise [17]. Unequal training budgets and hidden-target information in the simulator limit the comparison. These results cannot be transferred to the deployed LinUCB agent or to human physiology.

The integrated conclusion from these results is deliberately narrower than a product claim. The project has credible component research, meaningful negative findings and an implemented path for safe adaptation. It still needs a complete physical-device trace and a controlled participant study to establish the benefit of the combined experience.

6.4 Stakeholder and User Validation

An integrated participant study is proposed, not reported as completed. Recruitment must follow institutional approval and an informed-consent process. Participants should understand what is recorded, the non-diagnostic nature of predictions, potential VR discomfort, the role of cloud audio generation and their right to withdraw [24].

The initial study should prioritize feasibility, comfort and data validity. The sample size must be justified before recruitment using the primary outcome, expected paired variability, practical attrition and an appropriate power or precision argument. Selecting a convenient number and later presenting it as a powered clinical trial would be inappropriate.

Three package-level conditions can compare the complete experience: personalized static visuals and audio, rule-based adaptive visuals and audio, and contextual-bandit adaptive visuals and audio. The same supported scene, effective preference initialization, session schedule and retained generated content should be used wherever the research question requires content to be controlled.

Package-level improvement cannot identify which controller caused it. Separate component-isolation sessions should therefore hold audio static while comparing visual control, and hold visuals static while comparing audio control. A later factorial or serialized-action study can investigate interaction effects once attribution and burden are manageable.

Figure 6.2 outlines the proposed study sequence. Condition order should be counterbalanced, recovery periods declared and pre-session context recorded. One condition should be assigned per research session rather than changed casually through a dashboard while treating the result as a controlled experiment.

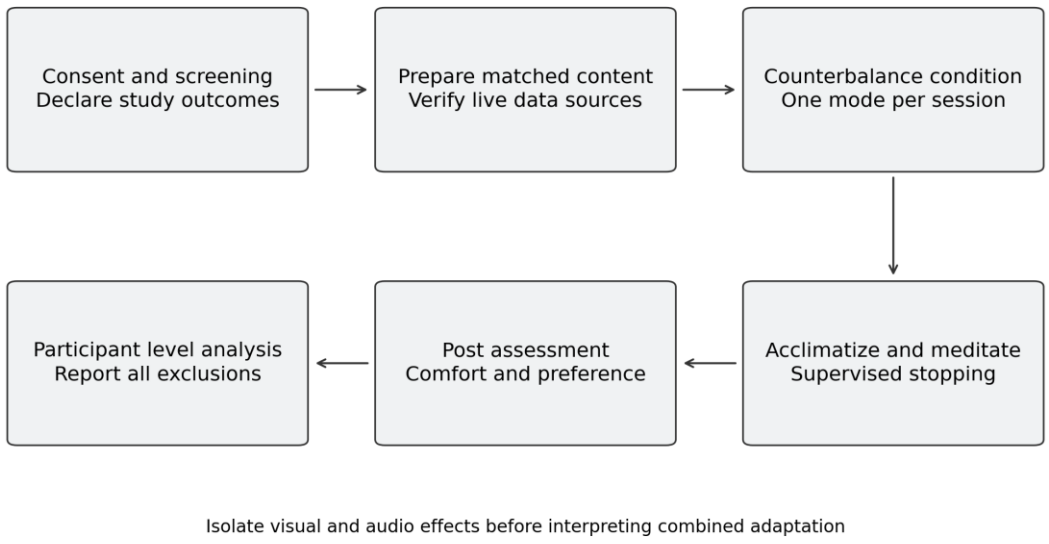


Figure 6.2 Proposed controlled participant evaluation workflow

The primary participant outcomes should be declared before data collection. Suitable feasibility measures include completion, discomfort, stoppage and usable-window rates. Preference satisfaction, naturalness, perceived personalization and non-

distraction can be recorded on clearly anchored scales. Pre/post subjective stress is complementary to physiology rather than a substitute for it.

Physiological analyses should use qualified non-overlapping or explicitly modelled windows. Voice pre/post change should retain recording quality and language. A lower voice score or stress proxy after meditation is an observed association unless the design and analysis support a stronger causal conclusion.

Analysis should respect the participant as the primary independent unit. A person producing many decisions must not dominate the study mean merely because their stream yielded more valid windows. The participant-level mean is formed first, followed by the mean and uncertainty across participants.

$$m_i = (1 / n_i) \Sigma_j r_{ij}$$

$$M = (1 / N) \Sigma_i m_i$$

Here n is the number of eligible outcomes for one participant and N is the number of participants with an analyzable outcome. Missing outcomes are not zero. A single participant does not provide a meaningful between-participant standard deviation. Mixed-effects or paired analyses may be appropriate once the final design and assumptions are declared.

6.5 Achievement of Common Objectives

Table 6.5 summarizes current achievement without treating source code as a substitute for empirical evidence. The status describes the reviewed snapshot, not a guarantee about every local installation or future deployment.

Table 6.5 Objective achievement at the reviewed snapshot

Objective	Current evidence	Remaining requirement
O1 Sensing to state	Firmware, mobile framing and B contracts implemented	Physical continuity, transfer and source audit
O2 Personalized session	Preferences, relay, scene and audio paths implemented	Full-device start and content-readiness trace
O3 Safe adaptation	Bounded online agents and technical checks	Approved settings and integrated fault tests
O4 Voice assessment	Recording path, inference and component experiments	Disable mock HRV and validate same-session joins
O5 Reproducibility	Versioned contracts and finalized visual export	Unified audio join and complete artefact manifest
O6 User benefit	Study design specified	Approval, recruitment, analysis and reporting

The major integration achievement is a coherent control-and-evidence architecture. Its limitations are operationally actionable: verify live inputs, approve research settings, join logs explicitly and isolate competing changes. These are stronger next steps than simply collecting more model training iterations without fixing the experiment.

6.6 Integrated Limitations and Threats

Construct validity is limited by the meaning of stress proxies. HRV-derived categories, voice valence and participant self-report measure different aspects of experience. None should be renamed a direct measurement of mental state. Their agreement is informative but does not establish ground truth.

Internal validity is threatened by time effects, respiration, novelty, expectancy, audio content and interacting controllers. A participant may improve during a quiet session regardless of adaptation. Content reuse, counterbalancing, component isolation and valid timing reduce these threats but do not eliminate them automatically.

External validity is limited by the current scene, sensor, device configuration, voice speaker pool and study population. A successful Japanese temple session does not establish performance in every virtual environment, on every wrist or across all languages. The short B recordings also cannot identify a full daily circadian rhythm reliably.

Statistical validity is threatened by overlapping windows, clip-level resampling, model selection on small datasets and unequal simulation budgets. Repeated seeds quantify training variability, not participant diversity. Pooled window scores should not be presented as equivalent to subject-held-out estimates.

Operational validity is limited by the global latest-stress interface, current mobile mock-HRV setting, absent live firmware temperature and unresolved approval configuration. These are explicit release blockers for certain claims. They are not evidence that the entire architecture is unusable, but they must be corrected or honestly bounded before a study.

Finally, online adaptation can change the policy during the experiment. Training and frozen evaluation phases should be distinguished, participant model versions saved and exploration settings recorded. Otherwise, an apparent condition difference may reflect different learning exposure rather than the intended controller comparison.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

The proposed value is a personalized immersive meditation experience that remains sensitive to uncertainty and participant control. Its distinctive system-level feature is not merely generated music or a stress score. It is the combination of preference initialization, bounded adaptation and an evidence trail explaining when the system chose to act or not act.

Near-term beneficiaries are research laboratories, university wellness researchers and supervised pilot programmes. These settings can evaluate comfort, feasibility and protocol adherence while retaining expert oversight. Positioning the prototype as an autonomous treatment product would exceed its current evidence.

The participant-facing benefit must be tested rather than presumed. A static personalized session might offer most of the value at lower complexity. If adaptive control does not improve comfort or preference satisfaction, that result should inform the design and the commercial proposition.

7.2 Market Alternatives and Differentiation

Relevant alternative categories include static meditation recordings, immersive relaxation environments, wearable biofeedback tools and conversational wellness applications. This report compares categories rather than claiming an exhaustive current market survey or inventing competitor performance.

Table 7.1 identifies the intended differentiation and the corresponding evidence burden. The table describes design intent, not proven market superiority. A procurement decision would require a separately maintained comparison of real products, prices, licences and supported devices.

Table 7.1 Alternative categories and intended differentiation

Alternative category	Typical emphasis	Laminar VR research distinction
Static audio meditation	Consistent low-complexity content	Preference-conditioned generation and bounded playback
Static VR relaxation	Immersion and scene design	Online visual and audio control with eligibility
Wearable biofeedback	Physiological awareness	Confidence and quality preserved through action selection
Voice wellness interface	Conversation and vocal interaction	Acoustic assessment separated from dialogue generation
General adaptive system	Response to model outputs	Applied-action and delayed-outcome audit trail

Differentiation creates additional responsibility. More data channels increase privacy exposure, and more adaptive mechanisms increase attribution complexity. These costs must be justified by demonstrated user value, not by the number of models in the architecture.

7.3 Business Model and Feasibility

A plausible early model is a supervised institutional pilot or research licence with support for deployment and analysis. A consumer subscription would require a much stronger reliability, safety and support foundation. Neither model is represented as an existing customer relationship or revenue stream.

Feasibility has hardware, operational and service dimensions. Hardware includes the headset, phone and wearable. Operations include installation, calibration, supervision, replacement parts and participant support. Services include generated audio, optional speech services, hosting and secure storage. Actual costs should be measured using the selected provider terms and observed usage.

A useful cost model separates one-time equipment from per-session usage. The main variable drivers are generated audio duration, retries, voice processing, storage

retention and staff time. Caching and reusing a participant-approved clip may reduce repeated generation, subject to provider permissions and the experimental design.

There is no defensible cost-per-relaxed-user figure before user outcomes and operating traces exist. A production feasibility study should report median and tail latency, failed requests, generation retries, support burden and session completion, not only the nominal API price.

7.4 Adoption Intellectual Property and Compliance

Adoption requires clear separation between a research wellness prototype and a regulated medical claim. Intended use, advertising language and decision consequences determine the appropriate review. The present system should not diagnose stress disorders, recommend treatment or replace professional care.

Privacy review should address physiological traces, voice recordings, derived affect, account identity, cross-border services, retention and deletion. Sri Lanka's data-protection framework and applicable amendments require formal institutional and legal review; the presence of consent screens or row-level database policies is not a complete compliance assessment [25].

Consent must explain the data actually used. If audio generation sends a preference prompt to an external provider, that transfer should be disclosed. If voice recordings are retained for model improvement, that purpose should be distinguished from immediate session assessment. Withdrawal and deletion procedures should be testable.

Intellectual property includes original integration code, model configuration, collected recordings and generated content. The team must review dataset, model, package and provider licences separately. WESAD's research-use conditions are especially relevant to any move beyond the academic prototype [4]. Ownership of jointly developed hardware and integration work should be agreed explicitly.

Participant research must follow institutional approval and relevant ethical principles [24]. No approval identifier is supplied in this draft. Before external use, a

documented risk assessment, incident procedure, software release process and security review are required.

7.5 Sustainable Development Goal Contribution

The project contributes to SDG 3 as a research effort exploring mental-well-being support, not as a demonstrated population intervention [23]. Its most credible near-term contribution is a transparent method for studying whether personalization is useful without suppressing uncertainty or negative results.

Environmental sustainability can be supported by repairable wearable construction, responsible battery handling, reusable audio assets and avoidance of unnecessary cloud requests. The physical prototype still needs verified electrical protection and maintenance procedures. A rechargeable battery alone does not make a device environmentally safe.

Social sustainability requires comfortable participation, accessible language, non-coercive consent and honest descriptions of capability. Reporting small-sample limitations and failed model adaptation is part of responsible innovation. It prevents premature deployment based on an attractive demonstration.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

Laminar VR brings together a jointly developed wearable and four individual research components within a common meditation workflow. Component B supplies uncertain physiological state estimates. Component C personalizes bounded visual changes. Component D contributes recorded-voice assessment and cross-modal interpretation. Component E personalizes bounded audio playback around scene- and preference-conditioned generated content.

The integrated architecture preserves the distinctions that matter scientifically: quality versus confidence, generation versus control, proposed versus applied action, fresh arrival versus valid post-action evidence and component experiments versus participant outcomes. These distinctions make the system auditable and provide a defensible basis for further research.

The reviewed evidence supports substantial implementation and selected technical verification. Fresh tests passed for relay contracts and audio analysis. Component B and D studies provide useful quantitative findings with explicit limitations. The evidence does not yet establish integrated physiological effectiveness, broad multilingual performance or production readiness.

The common project objective is therefore partially achieved at the research-prototype stage. The architecture and core paths are present; the final contribution must be completed through approved physical-device and participant validation. This conclusion recognises engineering progress without converting missing evidence into a favourable result.

8.2 Principal Integrated Contribution

The principal common contribution is an uncertainty-aware adaptation-and-evidence workflow. It connects acquisition, interpretation, bounded action and delayed observation across independent services. It makes safe inaction a valid outcome and gives participant interruption priority over model learning.

The two agents make personalization testable through bounded actions and traceable updates, but require controlled attribution. Voice provides complementary evidence without making disagreement a diagnosis. Separating generated content from playback control further supports interpretable experiments. None of these mechanisms alone guarantees physiological improvement.

8.3 Future Directions

The immediate priority is a reproducible live trace. Verify wearable timing and temperature provenance, disable mock HRV, confirm approved scene settings and bind all records to the same participant session before making integrated real-data claims.

Next, conduct a supervised feasibility study with declared conditions, valid windows and component isolation. Measure discomfort, completion, usable evidence, preference satisfaction and naturalness. Retain negative outcomes and analyse participants rather than treating overlapping windows as independent people.

Longitudinal data should support prospective sensor-transfer assessment for B and speaker-held-out multilingual validation for D. C and E require sufficient reliable outcomes to evaluate learning stability, action utility and frozen versus adapting policies.

A later study should introduce explicit action-attribution coordination or a factorial protocol. Test whether visual and audio adaptation are complementary, redundant or distracting rather than assuming that two useful agents improve the combined experience.

Production requires session-isolated transport, secure deployment, managed updates, approved signing, retention, deletion, monitoring, recovery, accessibility and verified headset performance. Continuous generation can be added when access and measured benefit justify it; real-time playback adaptation does not require continuous composition.

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APPENDIX A INTEGRATED INTERFACE SPECIFICATION

A.1 Acquisition and Physiological Envelope

The current BLE device name is WearableHealthMonitor. The reviewed service UUID is 7c69f001-7f70-4b0a-9c91-93d7f91b1001. The telemetry and raw characteristics use the same UUID family with f002 and f003 respectively. Older screenshots with another UUID must not be used as the firmware contract [13].

The raw packet is 41 bytes: one count byte followed by five little-endian pairs of unsigned uptime and infrared values. The phone should reject inconsistent length or count and should preserve discontinuity rather than fill arbitrary samples. Telemetry includes t, ir, red, noiseAvg, noisePeak, temp and flags; temp is currently null.

The following JSON is an illustrative stream contract, not a recording from a participant. The confidence value follows the top-two probability margin, 0.54 minus 0.36, rather than repeating the highest probability. HR is in beats per minute and RMSSD/SDNN are in milliseconds.

```
{
  "timestamp": 1787282898.4,
  "heartRate": 78.4,
  "rmssd": 34.1,
  "sdnn": 42.0,
  "stress": {
    "mode": "band",
    "level_low": 1,
    "level_high": 2,
    "label": "mild-to-moderate",
    "confidence": 0.18,
    "adjacent": true,
    "probabilities": {
      "relaxed": 0.08,
      "mild": 0.36,
      "moderate": 0.54,
      "high": 0.02
    },
    "continuous_score": 1.50
  },
  "signalQuality": 0.92,
  "windowStart": 1787282838.4,
  "windowEnd": 1787282898.4
}
```


GET /stress/latest returns the top-level prediction, not the nested stream envelope. It is unavailable before the first valid window. Consumers must not invent HRV values absent from that endpoint [15], [19].

A.2 Session and Evidence Contracts

The relay's session schema is Laminar-session-v1. Pairing uses a one-time six-digit code with an approximately five-minute lifetime. A paired Quest receives the configured session and command stream. Terminal visual-log retrieval and acknowledgement are separate operations [21].

Table A.1 lists the minimum fields for a research export. Some fields are stored across existing files rather than in one unified export. Completing an explicit common join is a remaining integration task, not an assumed property of the current logs.

Table A.1 Minimum integrated research export fields

Record group	Required context	Validation
Session	Pseudonymous participant, relay ID, condition and scene	One declared condition and version
Acquisition	Device, rate, source, gaps and time anchor	No fabricated samples or temperature
State	Window boundaries, distribution, confidence and quality	Finite values and valid provenance
Decision	Agent, model version, context and chosen action	Eligible and correctly attributed
Application	Safe target, actual output and completion time	Proposed action matches applied change
Outcome	Post window, reward terms or cancellation	Invalid outcomes remain missing
Assessment	Voice phase, recording quality and self-report	Same participant and session

APPENDIX B TEAM CONTRIBUTION DECLARATION

The group should confirm this declaration before submission. Component A was developed jointly by all four members. Table B.1 identifies the individual responsibilities and Table B.2 provides the contribution confirmation. Personal details and signatures remain blank until the team agrees that the descriptions are accurate.

Table B.1 Member identification and individual responsibility

Member name	Student ID	Individual component
_____	_____	B Physiological inference
_____	_____	C Adaptive visual environment
_____	_____	D Voice based assessment
_____	_____	E Adaptive audio control

Joint work includes wearable prototyping and the agreed common integration tasks. Individual model development, experiment execution and interpretation should be supported by the relevant source history, notebooks and test records. This draft does not allocate unsupported effort percentages.

Table B.2 Contribution confirmation

Member	Signature	Date
Member B	_____	_____
Member C	_____	_____
Member D	_____	_____
Member E	_____	_____

APPENDIX C PROPOSED PARTICIPANT STUDY RECORD

C.1 Session Preparation and Stopping

Before enrolment, record the ethics approval reference, approved information sheet, consent version, eligibility criteria, withdrawal process and primary outcome. The participant count and analysis plan must be justified and registered before data collection. No approval or sample size is supplied by this draft.

Before each session, verify the wearable connection, raw signal continuity, declared temperature source, B model readiness, absence of mock HRV, paired headset, generated audio readiness, comfortable starting volume and approved scene settings. Record the assigned condition and model versions before adaptation begins.

Explain pause and stop controls in ordinary language. The operator should stop the session for participant request, nausea, headache, eye strain, unexpected audio discomfort, equipment fault or a protocol-defined safety event. Record the reason and do not treat an aborted session as a completed low-reward session.

C.2 Suggested Feedback and Analysis Record

Use the institution-approved questionnaire wording and scoring rules. Suggested session-specific items ask whether changes were natural, distracting, comfortable and consistent with preferences. The participant should also be able to describe unwanted transitions, sound levels and motion in their own words.

Record pre/post subjective state, valid physiological coverage, recording quality and the reason for every excluded outcome. Retain condition order and recovery duration. Report the number of participants, sessions, clips and windows separately because these are not interchangeable sample sizes.

The final analysis should report estimates and uncertainty, not only favourable averages. Discomfort, failed sessions and missing windows belong in the results. Any exploratory analysis not specified in advance should be labelled exploratory rather than presented as a confirmed primary finding.

APPENDIX D REPRODUCIBILITY AND SOURCE REGISTER

D.1 Evidence Provenance

The common-report source snapshot was reviewed on 11 October 2026. Table D.1 identifies the principal material used. The supplied summaries were compared with runtime code when describing present behaviour. A source inspection is not a reproduction of model training or a physical-device trial.

Table D.1 Principal evidence sources

Source group	Material	Evidence class
Common report requirements	Supplied common DOCX and formatting PDF	Structure and presentation authority
Shared wearable	Supplied wearable MD and firmware	Design description plus code inspection
B research	Supplied master MD and component-b sources	Reported experiments plus configuration
C research	Learning contract, Unity sources and validation docs	Implementation and historical tests
D research	Supplied submission, ablation and handoff docs	Reported experiments plus code inspection
E research	Audio architecture, Unity agent and analysis tests	Implementation and fresh analysis tests
Common integration	Mobile, relay, launcher and relay tests	Implementation and fresh in-process tests

Fresh Python unittest verification passed six relay tests and five audio-analysis tests. It did not start paid generation requests or train new B or D models. The reproducible commands appear in Section D.2.

D.2 Experiment Manifest Requirements

A reproducible participant release should retain the repository revision, dirty-change manifest, dependency versions, firmware build, mobile APK signature, Unity build, model hashes, provider model name and every controller configuration. It should also retain the approved protocol and the exact analysis command.

The following PowerShell commands reproduce the fresh verification scopes from the common repository. The first command uses line continuation to preserve both relay modules in one test invocation.

```
python -m unittest `
  services.session_relay.test_session_store `
  services.session_relay.test_relay_app -v
python -m unittest discover -s tests -p test_audio_online_evaluation.py -v
```

For online agents, save the participant model state before and after each session. Declare whether learning and exploration were enabled. Keep simulated and live sources separate. For generated content, retain the prompt version, scene reference, request identity, generation outcome and actual clip used, subject to licence and privacy requirements.

Before publishing, remove credentials and direct identifiers from logs and examples. Keep access-controlled originals where required by the approved retention plan. A public reproducibility package may use synthetic interface fixtures, but it must label them as synthetic rather than suggesting that they are anonymized participant results.

APPENDIX E RESEARCH AND PRODUCTION READINESS GATES

Table E.1 converts the report's limitations into release gates. Passing these gates requires recorded evidence and a responsible reviewer. A deadline or successful demonstration does not substitute for approval.

Table E.1 Prioritized readiness gates

Gate	Required evidence	Release consequence
Live input provenance	Physical PPG trace, temperature declaration and no mock HRV	Required for real-data claims
Approved research settings	Signed protocol and reconciled scene approval flag	Required before participant use
Identity and attribution	Common session join and uncontaminated outcomes	Required for interpretable learning results
Device stability	Full session on phone, wearable and headset	Required for supervised pilot readiness
Participant evaluation	Approved study and declared analysis	Required for benefit claims
Security and privacy	TLS, isolation, signing, retention and review	Required for external deployment
Product operation	Monitoring, support, recovery and accessibility	Required for sustainable service

The gates are intentionally different. A supervised research pilot can tolerate documented development constraints that would be unacceptable in a public service. Conversely, a secure production deployment does not prove that the intervention is effective. Technical, scientific and operational readiness should be assessed independently.

APPENDIX F ASSISTANCE AND AUTHOR VERIFICATION

Generative AI assisted with source synthesis, report drafting, diagram preparation and document formatting. Automated tools supported selected test execution and layout checks. This assistance did not recruit participants, grant ethics approval, conduct the reported component training studies or establish clinical effectiveness.

The four authors remain responsible for checking the technical descriptions, numerical results, citations, licences, contribution allocation and compliance with university policy. Before submission, they should complete the blank identity fields, confirm the declaration, verify all supplied component results against original experiment records and obtain supervisor review.

The original source template and formatting document were retained unchanged. This common thesis is a draft for author verification and institutional review. Its limitations are part of the research record, not instructions to omit unfavourable evidence from the final submission.